

App Name: Corona Supervisor

Overview

A bilingual (Arabic/English) mobile field sales supervision app for FMCG companies. Field supervisors use it to log customer visits, perform shelf audits, capture product photos, record competitor products, and place orders — all tied to their GPS location. Managers get real-time visibility into field activity.

Tech Stack

- **Backend & Database:** Supabase (PostgreSQL)
- **Auth:** Supabase Auth (email + password)
- **File Storage:** Supabase Storage (photos)
- **AI:** Claude Vision API (shelf image analysis)
- **Language:** Arabic UI (RTL), English variable names in code

User Roles

1. **Supervisor** — field agent using the mobile app
2. **Manager** — admin with full visibility (web dashboard, future phase)

Database Tables (Supabase)

```
sql

supervisors      (id, full_name, phone, branch, role, is_active)
customers        (id, name, type, latitude, longitude, address)
visits           (id, supervisor_id, customer_id, status, latitude, longitude, geo_di
products         (id, name, category, image_url, is_active)
shelf_audit      (id, visit_id, product_id, is_present, quantity, photo_url, ai_analy
competitor_products (id, visit_id, brand_name, product_name, quantity, photo_url)
orders           (id, visit_id, supervisor_id, customer_id, product_id, quantity, sta
targets          (id, supervisor_id, target_date, visits_target, audit_target)
planograms       (id, category, image_url, description)
```

Screens & Logic

1. Splash & Login Screen

- App logo centered
 - Login with email + password via Supabase Auth
 - On success → check `supervisors.is_active`, if false show "الحساب غير مفعل"
 - Remember session (stay logged in)
-

2. Home Screen

- Header: "مرحباً [supervisor name] 🙌"
 - **Target Card:** progress bar showing today's visits done vs target (from `targets` table)
 - **Quick Actions:**
 - 📍 زيارة جديدة
 - 📅 زياراتي اليوم
 - 📊 إحصائياتي
 - 🗺️ مسار اليوم
 - **Smart Notifications strip** at top (unread count badge)
-

3. New Visit Screen (Location)

- Button "أرسل موقعك" 📍 → get device GPS coordinates
- Calculate distance to all customers using Haversine formula
- Show nearest 5 customers as tappable cards:
 - Customer name, type, distance (meters if <1km, else km)
- Tap a customer → go to Customer Visit Screen

Haversine function (use in code):

```
js
```

```
function distanceKm(lat1, lon1, lat2, lon2) {
  const R = 6371;
  const dLat = (lat2 - lat1) * Math.PI / 180;
  const dLon = (lon2 - lon1) * Math.PI / 180;
  const a = Math.sin(dLat/2)**2 +
    Math.cos(lat1 * Math.PI/180) *
    Math.cos(lat2 * Math.PI/180) *
    Math.sin(dLon/2)**2;
  return R * 2 * Math.atan2(Math.sqrt(a), Math.sqrt(1 - a));
}
```

4. Customer Visit Screen

- Customer name, type, address at top
- **Geofence check:** if distance > 500m → show warning "م:أنت خارج النطاق الجغرافي — المسافة —"
- **Duplicate check:** if same customer visited in last 30 min → show warning
- Visit status buttons:
 -  متعامل
 -  غير متعامل
 -  غير موجود
- After selecting status → save to **visits** table → show action menu:
 -  كشف الرف
 -  منتجات المنافسين
 -  تسجيل أو ردر
 -  إنهاء الزيارة

5. Shelf Audit Screen ★

- Fetch all active products from **products** table
- Display as scrollable list, each product card has:
 - Product name + category
 - Toggle:  موجود  / غير موجود
 - If موجود → show quantity input (number)
 -  "التقط صورة" button → open camera
 - After photo taken → auto-send to **Claude Vision API**

- Show AI result badge: quantity estimate + "مرتّب/غير مرتّب"
- Bottom: "إرسال الكشف  button"
 - Upload all photos to Supabase Storage
 - Save all rows to `shelf_audit` table
 - Show success toast "تم حفظ كشف الرف بنجاح"

Claude Vision API call:

```
js
const response = await fetch("https://api.anthropic.com/v1/messages", {
  method: "POST",
  headers: { "Content-Type": "application/json" },
  body: JSON.stringify({
    model: "claude-sonnet-4-6",
    max_tokens: 300,
    messages: [{
      role: "user",
      content: [
        { type: "image", source: { type: "base64", media_type: "image/jpeg", data: b
        { type: "text", text: "فقط JSON حل صورة الرف. أجب بـ"
      ]
    }]
  })
});
```

6. Competitor Products Screen

- Header: "منتجات المنافسين 
- Add button → form with:
 - Brand name (text)
 - Product name (text)
 - Quantity (number)
 -  صورة (optional)
- List of added competitor products for this visit
- Save to `competitor_products` table

7. Order Screen

- Fetch products from `products` table
 - Searchable list with product name + category
 - Tap product → input quantity
 - Cart summary at bottom showing selected items + total units
 - "✅ إرسال الأوردر" → save to `orders` table with status = 'pending'
 - Success message: "تم تسجيل الأوردر بنجاح" ✅
-

8. My Visits Today Screen

- Filter `visits` by supervisor_id + today's date
 - Show list with:
 - Customer name
 - Status icon (● ● ●)
 - Time of visit
 - Has shelf audit? (📦 badge)
 - Total count at top
-

9. My Stats Screen

- Weekly stats from `visits` table:
 - Total visits this week
 - متعامل count ●
 - غير متعامل count ●
 - غير موجود count ●
 - Compliance % = $\frac{\text{متعامل}}{\text{total}} \times 100$
- Target vs Actual bar chart (this week)
- **Customer Health Score** for each visited customer:

Score = visit_frequency(30%) + product_coverage(40%) + order_regularity(30%)

● 80-100 | ● 60-79 | ● 0-59

10. Route Optimization Screen

- Fetch today's target customers from `targets` + `customers`
 - Calculate optimal route using Nearest Neighbor algorithm starting from current location
 - Display ordered list with estimated distances
 - "افتح في الخريطة" button per customer → opens Google Maps directions
-

11. Notifications Screen

- Smart alerts stored locally + from Supabase:
 - "لم يُزَر منذ 7 أيام [X] عميل"
 - "[Z] غير موجود 3 زيارات متتالية عند [Y] صنف"
 - "عملاء في نطاق 500م [N] لديك"
 - "زيارة لإتمام هدف اليوم [N] تبقى" (after 2pm if target not met)
-

UI/UX Guidelines

- **RTL layout** throughout
- **Color scheme:** Primary `#1A56DB` (blue), Success `#10B981` (green), Danger `#EF4444` (red)
- **Font:** Cairo (Arabic Google Font)
- Bottom navigation bar with 4 tabs: الرئيسية | زيارة | تقارير | إشعارات
- Cards with subtle shadows, rounded corners (12px)
- Loading skeletons while fetching data
- Offline-friendly: queue failed saves and retry when connection restored